

✓ Employee Management System

Developed By

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Technologies Used

- Python
- SQLite
- Pandas
- Google Colab

Features

- Add Employee
- View Employees
- Search Employee
- Update Employee
- Delete Employee

Creating Database

```
1 import sqlite3
2
3 conn = sqlite3.connect("employee_management.db")
4
5 cursor = conn.cursor()
6
7 cursor.execute("""
8 CREATE TABLE IF NOT EXISTS employees(
9     employee_id INTEGER PRIMARY KEY AUTOINCREMENT,
10    full_name TEXT,
11    department TEXT,
12    job_title TEXT,
13    email TEXT,
14    phone TEXT,
15    salary REAL,
16    joining_date TEXT
17 )
18 """)
19
20 conn.commit()
21 conn.close()
22
23 print("Database Created Successfully")
```

Database Created Successfully

Adding Employee

```
1 import sqlite3
2
3 conn = sqlite3.connect("employee_management.db")
4 cursor = conn.cursor()
5
6 cursor.execute("""
7 INSERT INTO employees
8 (full_name, department, job_title, email, phone, salary, joining_date)
9 VALUES
10 (
11 'Eric Rathod',
12 'IT',
13 'Developer',
```

```

14 'eric@test.com',
15 '6479619811',
16 65000,
17 '2026-01-01'
18 )
19 """)
20
21 conn.commit()
22 conn.close()
23

```

Employee Added Successfully

View Employees

```

1 import sqlite3
2 import pandas as pd
3
4 conn = sqlite3.connect("employee_management.db")
5
6 df = pd.read_sql_query(
7     "SELECT * FROM employees",
8     conn
9 )
10
11 conn.close()
12
13 df
14

```

	employee_id	full_name	department	job_title	email	phone	salary	joining_date	
0	1	Eric Rathod	IT	Developer	eric@test.com	6479619811	65000.0	2026-01-01	

Adding More Employees

```

1 import sqlite3
2
3 conn = sqlite3.connect("employee_management.db")
4 cursor = conn.cursor()
5
6 employees = [
7     ("Amit Patel", "HR", "HR Assistant", "amit@test.com", "6471112222", 52000, "2026-02-10"),
8     ("Sara Khan", "Finance", "Accountant", "sara@test.com", "6472223333", 58000, "2026-03-15"),
9     ("John Smith", "IT", "System Analyst", "john@test.com", "6473334444", 70000, "2026-04-01"),
10    ("Priya Sharma", "Operations", "Operations Coordinator", "priya@test.com", "6474445555", 56000, "2026-05-20")
11 ]
12
13 cursor.executemany("""
14 INSERT INTO employees
15 (full_name, department, job_title, email, phone, salary, joining_date)
16 VALUES (?, ?, ?, ?, ?, ?, ?)
17 """, employees)
18
19 conn.commit()
20 conn.close()
21
22 print("Multiple employees added successfully")

```

Multiple employees added successfully

Viewing Updated Table

```

1 import sqlite3
2 import pandas as pd

```

```
3
4 conn = sqlite3.connect("employee_management.db")
5
6 df = pd.read_sql_query("SELECT * FROM employees", conn)
7
8 conn.close()
9
10
```

	employee_id	full_name	department	job_title	email	phone	salary	joining_date	
0	1	Eric Rathod	IT	Developer	eric@test.com	6479619811	65000.0	2026-01-01	
1	2	Amit Patel	HR	HR Assistant	amit@test.com	6471112222	52000.0	2026-02-10	
2	3	Sara Khan	Finance	Accountant	sara@test.com	6472223333	58000.0	2026-03-15	
3	4	John Smith	IT	System Analyst	john@test.com	6473334444	70000.0	2026-04-01	
4	5	Priya Sharma	Operations	Operations Coordinator	priya@test.com	6474445555	56000.0	2026-05-20	

Next steps:

Generate code with df

New interactive sheet

Search Employee

```
1 import sqlite3
2 import pandas as pd
3
4 search_name = input("Enter employee name to search: ")
5
6 conn = sqlite3.connect("employee_management.db")
7
8 query = """
9 SELECT * FROM employees
10 WHERE full_name LIKE ?
11 """
12
13 df = pd.read_sql_query(query, conn, params=('%' + search_name + '%',))
14
15 conn.close()
16
17 df
```

Enter employee name to search: Eric

	employee_id	full_name	department	job_title	email	phone	salary	joining_date	
0	1	Eric Rathod	IT	Developer	eric@test.com	6479619811	65000.0	2026-01-01	

Update Employee

```
1 import sqlite3
2
3 employee_id = int(input("Enter Employee ID to update: "))
4 new_department = input("Enter new department: ")
5 new_job_title = input("Enter new job title: ")
6 new_salary = float(input("Enter new salary: "))
7
8 conn = sqlite3.connect("employee_management.db")
9 cursor = conn.cursor()
10
11 cursor.execute("""
12 UPDATE employees
13 SET department = ?, job_title = ?, salary = ?
14 WHERE employee_id = ?
15 """, (new_department, new_job_title, new_salary, employee_id))
16
17 conn.commit()
18 conn.close()
19
20 print("Employee Updated Successfully")
```

Enter Employee ID to update: 1
Enter new department: IT

Enter new job title: Junior Developer

Enter new salary: 68000

Employee Updated Successfully

Delete Employee

```
1 import sqlite3
2
3 employee_id = int(input("Enter Employee ID to delete: "))
4
5 conn = sqlite3.connect("employee_management.db")
6 cursor = conn.cursor()
7
8 cursor.execute("""
9 DELETE FROM employees
10 WHERE employee_id = ?
11 """, (employee_id,))
12
13 conn.commit()
14 conn.close()
15
16 print("Employee Deleted Successfully")
```

Enter Employee ID to delete: 5

Employee Deleted Successfully

Verify Deletion

```
1 import sqlite3
2 import pandas as pd
3
4 conn = sqlite3.connect("employee_management.db")
5
6 df = pd.read_sql_query(
7     "SELECT * FROM employees",
8     conn
9 )
10
11 conn.close()
12
13 df
```

	employee_id	full_name	department	job_title	email	phone	salary	joining_date	
0	1	Eric Rathod	IT	Junior Developer	eric@test.com	6479619811	68000.0	2026-01-01	
1	2	Amit Patel	HR	HR Assistant	amit@test.com	6471112222	52000.0	2026-02-10	
2	3	Sara Khan	Finance	Accountant	sara@test.com	6472223333	58000.0	2026-03-15	
3	4	John Smith	IT	System Analyst	john@test.com	6473334444	70000.0	2026-04-01	

Next steps:

Generate code with df

New interactive sheet

Department Report

```
1 import sqlite3
2 import pandas as pd
3
4 conn = sqlite3.connect("employee_management.db")
5
6 query = """
7 SELECT
8     department,
9     COUNT(*) AS Total_Employees,
10    ROUND(AVG(salary),2) AS Average_Salary
11 FROM employees
12 GROUP BY department
13 """
14
15 report = pd.read_sql_query(query, conn)
16
```

```
17 conn.close()
18
19 report
```

	department	Total_Employees	Average_Salary	
0	Finance	1	58000.0	 
1	HR	1	52000.0	
2	IT	2	69000.0	

Next steps:

Generate code with report

New interactive sheet